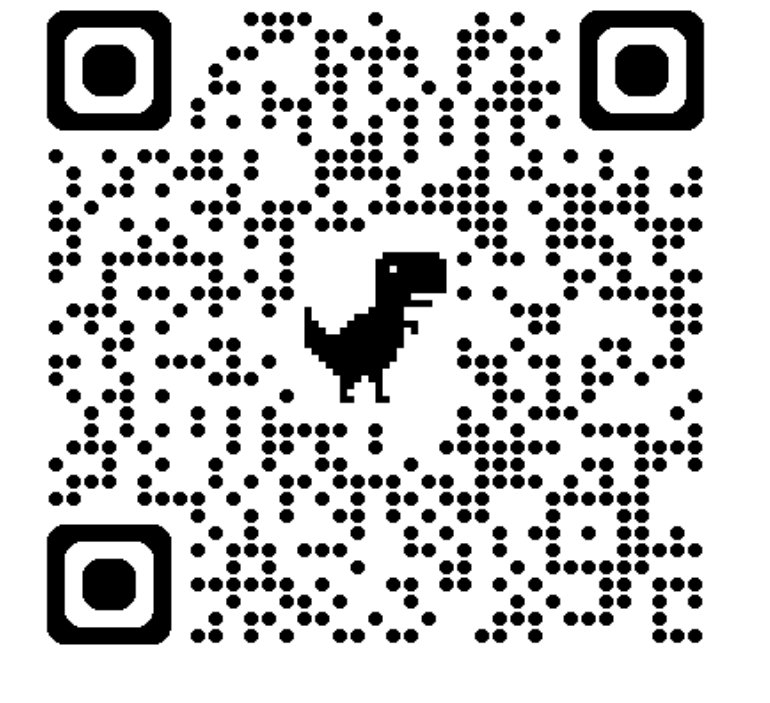




Introduction

- Salt-and-pepper noise severely degrades images by replacing pixels with extreme intensity values.
- Classical denoising methods often struggle at high noise densities, leading to residual noise or loss of structural details.
- We propose **I-HDF** (Iterative Hybrid Differential Filtering), combining a modified adaptive median filter with approximate differential operator-based iterative restoration.
- **Key Contributions:**
 - Hybrid framework integrating approximate differential operators with iterative least-squares (LS) optimization.
 - Robust high-density salt-and-pepper noise removal.
 - Edge and texture preservation.
 - Explainable, reproducible, and computationally efficient design.

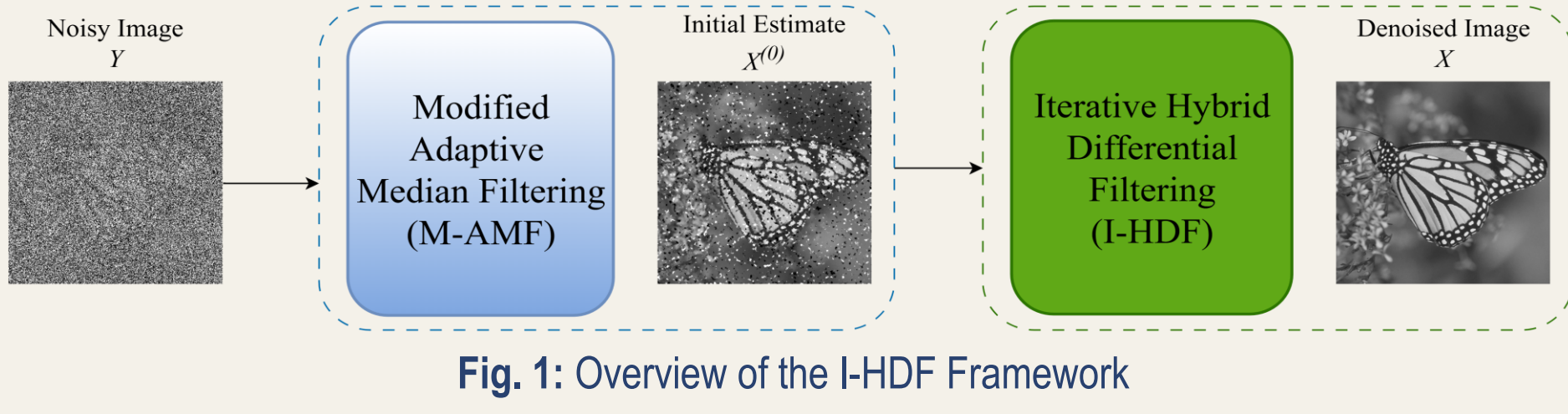


Fig. 1: Overview of the I-HDF Framework

Proposed Method: I-HDF

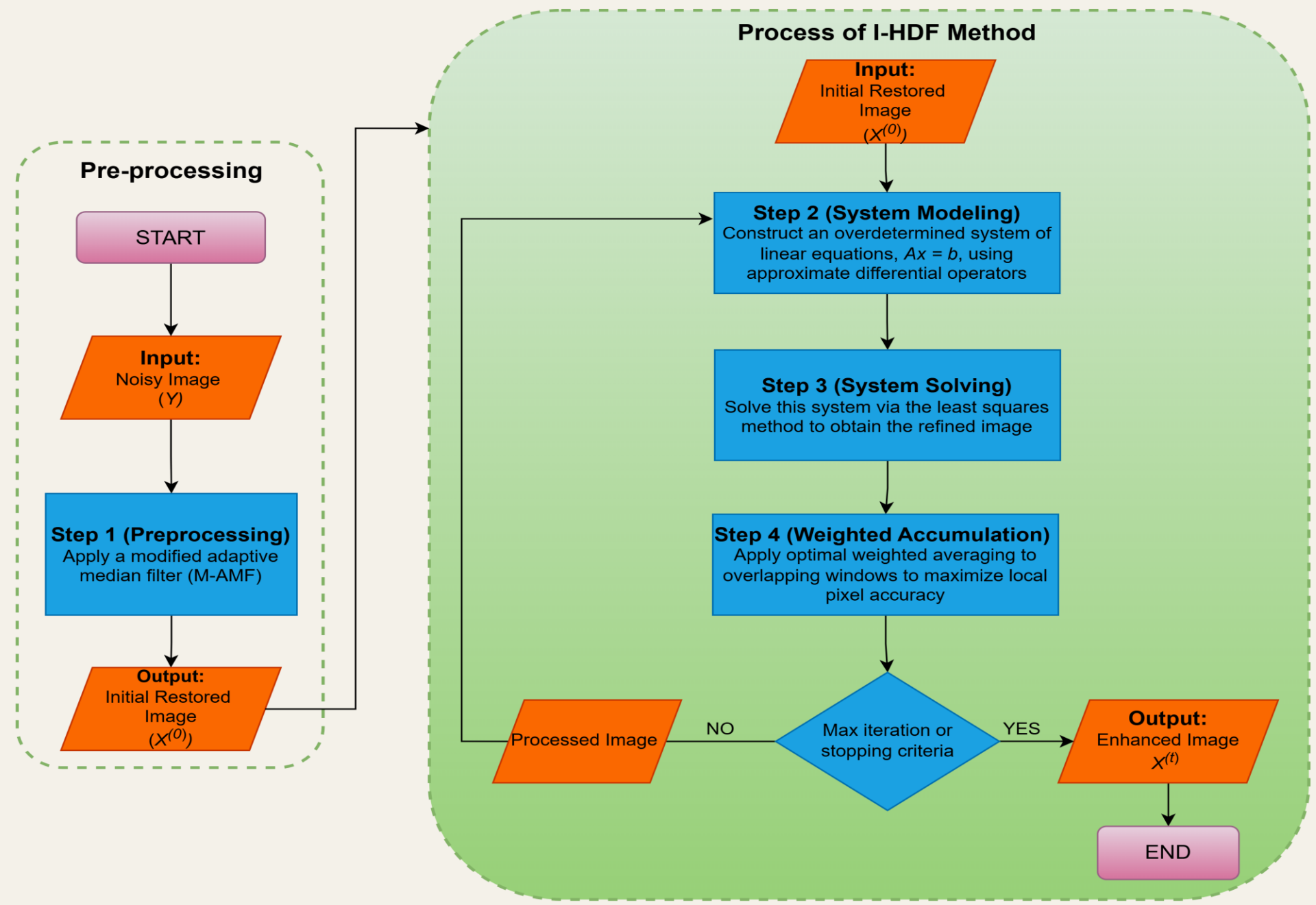


Fig. 2: Workflow of the Proposed I-HDF Method

Stage 1: Modified Adaptive Median Filtering (M-AMF)

- Identifies noisy pixels via extreme-value thresholding:
$$M = \{(i, j) | Y_{i,j} \in \{0, 255\}\}$$
- Generates a robust initial estimate $X^{(0)}$ from the noisy image $Y \in \mathbb{R}^{H \times W}$.
- Dynamically enlarges the local window to find reliable median values before differential refinement.

Stage 2: Iterative Hybrid Differential Filtering (I-HDF)

- Discretized with uniform pixel spacing ($h_x = h_y = 1$) via central difference, three-point forward and backward stencil formulas:

$$\delta_x^{(c)} f(x, y) = \frac{f(x + h_x, y) - f(x - h_x, y)}{2h_x} + O(h_x^2)$$

$$\delta_x^{(3+)} f(x, y) = \frac{-f(x + 2h_x, y) + 4f(x + h_x, y) - 3f(x, y)}{2h_x} + O(h_x^2)$$

$$\delta_x^{(3-)} f(x, y) = \frac{f(x - 2h_x, y) - 4f(x - h_x, y) + 3f(x, y)}{2h_x} + O(h_x^2)$$

The same operators are applied along both x - and y -directions.

- Stencil constraints:

$$\begin{aligned} \delta_x^{(c)} &= \delta_x^{(3+)}, & \delta_x^{(c)} &= \delta_x^{(3-)}, & \delta_x^{(3+)} &= \delta_x^{(3-)}, \\ \delta_y^{(c)} &= \delta_y^{(3+)}, & \delta_y^{(c)} &= \delta_y^{(3-)}, & \delta_y^{(3+)} &= \delta_y^{(3-)} \end{aligned}$$

These constraints define the discrete weight vectors:

$$W_{\text{EDGE}} = [-1, 3, -3, 1]$$

$$W_{\text{MID}} = [-1, 4, -6, 4, -1]$$

- Constructs an overdetermined linear system $Ax = b$ over overlapping 5×5 local windows $\Omega \subset X^{(t-1)}$:

$$\hat{x} = \arg \min_x \|Ax - b\|_2^2$$

- Aggregates overlapping patch estimates using the spatial reliability matrix W_ω :

$$W_\omega = \begin{bmatrix} 1 & 2 & 3 & 2 & 1 \\ 2 & 4 & 6 & 4 & 2 \\ 3 & 6 & 9 & 6 & 3 \\ 2 & 4 & 6 & 4 & 2 \\ 1 & 2 & 3 & 2 & 1 \end{bmatrix}$$

- Iteratively updates the reconstructed image until the relative change between consecutive iterations satisfies:

$$\delta^{(t)} = \frac{\|X^{(t)} - X^{(t-1)}\|_F^2}{\|X^{(t)}\|_F^2} < \varepsilon$$

Experimental Setup

- **Datasets:**
 - 12STI (12 images, 512×512),
 - BSD68 (68 images, 321×481 and 481×321),
 - Medical (10 images with different resolutions)
- **Noise Levels:** Salt-and-pepper noise with $d = \{0.2, 0.4, 0.6, 0.8\}$
- **Compared Methods:** MF, AMF, DAMF, HDMRMF, BM3D, NLM, and Wavelet denoising
- **Metrics:** PSNR, SSIM, and execution time
- **Hardware:** Intel Core i7-1255U (1.70 GHz), Windows 11

Experimental Results: 12STI & BSD68

Table 1: Comparative Analysis of PSNR (dB) and SSIM Results Across Different Datasets

Dataset Method		PSNR				SSIM			
		0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8
12STI	MF	29.02	26.66	18.51	10.18	0.8010	0.7647	0.4847	0.0733
	AMF	34.60	30.75	26.21	15.42	0.9474	0.9048	0.8173	0.3793
	DAMF	37.58	33.41	30.43	27.33	0.9751	0.9406	0.8909	0.8075
	HDMRMF	38.01	34.25	31.09	27.64	0.9767	0.9466	0.8997	0.8140
	I-HDF	41.24	37.07	33.63	29.65	0.9814	0.9560	0.9179	0.8453
	BM3D	22.02	20.45	17.58	15.26	0.4641	0.5214	0.4780	0.4140
	NLM	22.63	19.36	16.86	14.88	0.4572	0.4058	0.3516	0.2991
	Wavelet	17.71	18.58	16.24	14.46	0.2766	0.5061	0.4844	0.4707
BSD68	MF	25.42	23.91	17.62	9.90	0.7023	0.6672	0.4307	0.0738
	AMF	30.74	27.55	23.94	14.85	0.9265	0.8706	0.7680	0.3633
	DAMF	32.54	28.67	25.98	23.33	0.9558	0.9067	0.8379	0.7279
	HDMRMF	31.63	28.42	25.92	23.22	0.9571	0.9127	0.8470	0.7350
	I-HDF	35.11	30.83	27.67	24.59	0.9714	0.9305	0.8677	0.7610
	BM3D	20.68	19.31	16.71	14.49	0.4308	0.4352	0.3849	0.3341
	NLM	21.34	18.43	16.05	14.14	0.4097	0.3435	0.2897	0.2446
	Wavelet	16.57	17.84	15.67	14.03	0.2595	0.4188	0.3895	0.3746

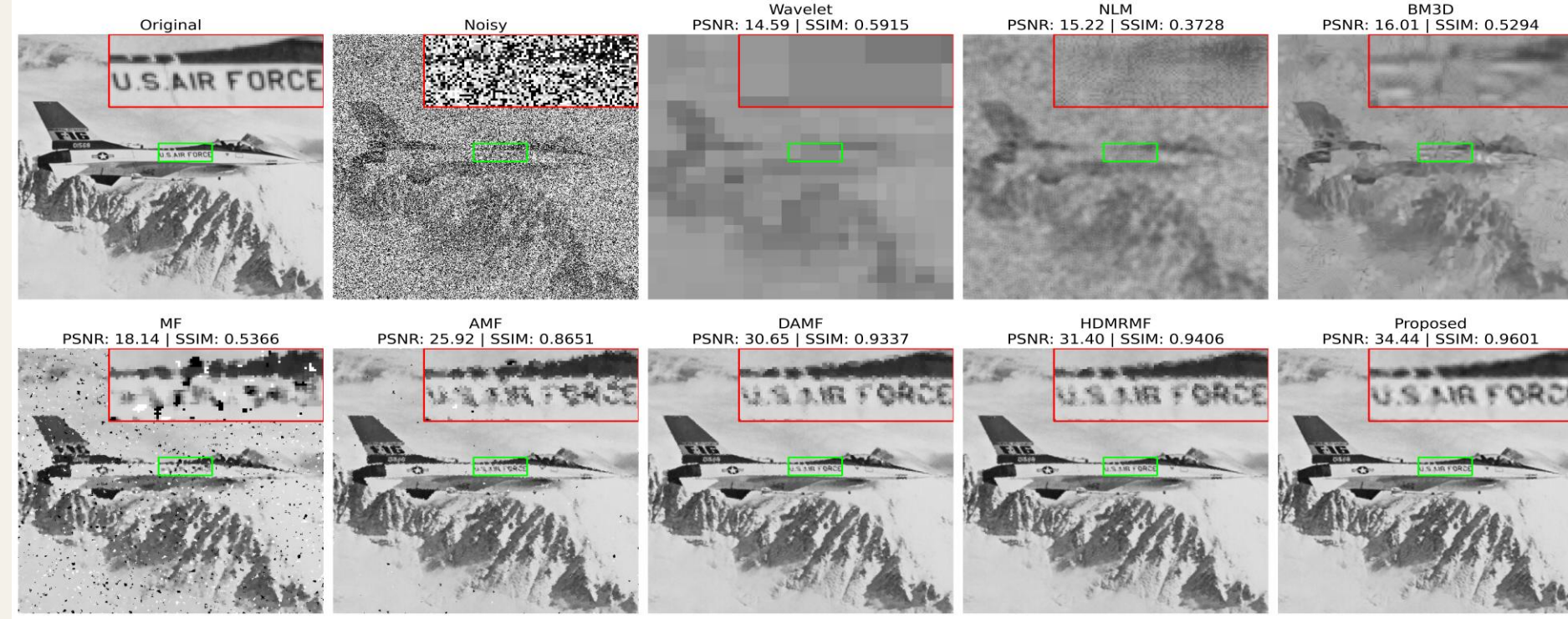


Fig. 3: Results for the Plane image under 0.6 noise
I-HDF: 34.44 dB | 0.9601
HDMRMF: 31.40 dB | 0.9406

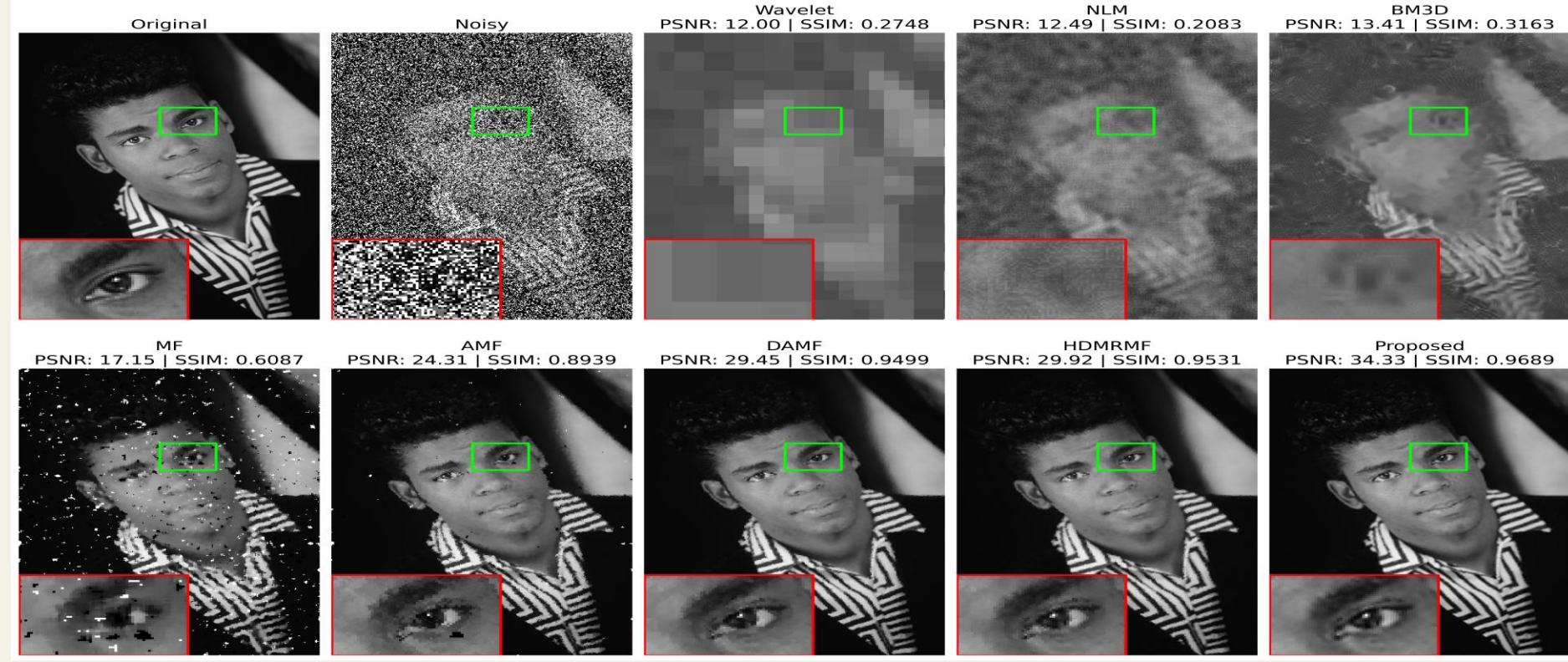


Fig. 4: Results for the Man image under 0.6 noise
I-HDF: 34.33 dB | 0.9689
HDMRMF: 29.92 dB | 0.9531

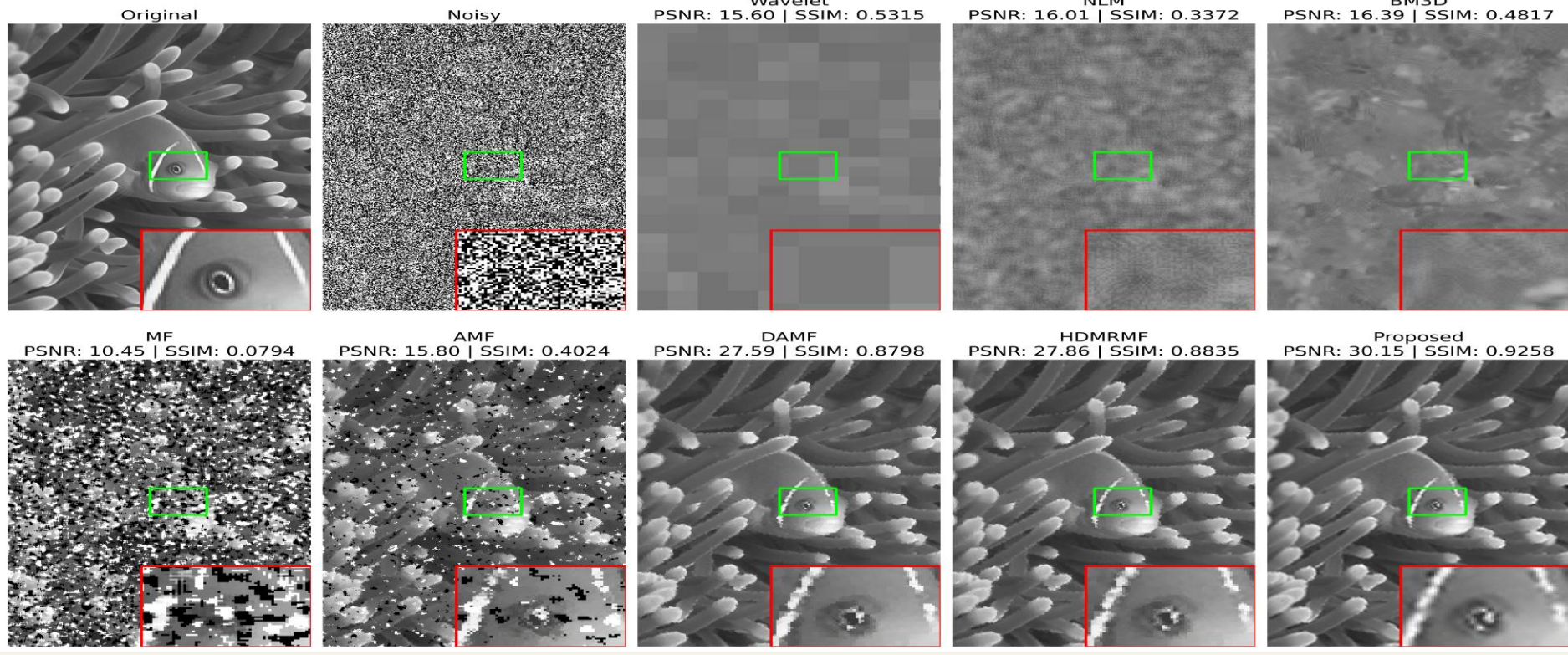


Fig. 5: Results for the Fish image under 0.8 noise
I-HDF: 30.15 dB | 0.9258
HDMRMF: 27.86 dB | 0.8835

- **Consistent Quantitative Superiority:** I-HDF consistently achieves the highest PSNR and SSIM scores across all noise densities ($d = 0.2 \rightarrow 0.8$), significantly surpassing both classical filters and recent methods like DAMF and HDMRMF.

- **Sharp Structural and Edge Recovery:** As evidenced by visual comparisons, the I-HDF method effectively prevents the over-smoothing and structural blurring typical of median-based filters, reliably preserving sharp high-frequency edges and fine texture patterns.

- **Robustness Under Extreme Corruption:** Even under severe noise conditions ($d = 0.8$), the proposed iterative hybrid differential scheme reliably recovers sharp contours and fine background details.

Real-World Assessment: Medical Dataset

Table 2: Comparative Analysis of PSNR (dB) and SSIM Results on Medical Dataset

Method	PSNR				SSIM			
	0.2	0.4	0.6	0.8	0.2	0.4	0.6	0.8
MF	33.34	29.07	18.43	9.64	0.9076	0.8722	0.5621	0.0843
AMF	38.94	34.12	27.76	15.05	0.9784	0.9499	0.8835	0.4317
DAMF	39.83	35.48	32.46	29.13	0.9819	0.9638	0.9358	0.8793
HDMRMF	39.18	35.86	32.87	29.39	0.9834	0.9696	0.9446	0.8882
I-HDF	47.43	43.32	39.59	34.38	0.9934	0.9854	0.9701	0.9304
BM3D	19.73	18.95	15.51	12.76	0.3688	0.4735	0.4348	0.3770
NLM	21.64	18.00	14.74	12.37	0.4014	0.3721	0.3187	0.2673
Wavelet	17.16	17.15	14.20	12.15	0.2377	0.4671	0.4313	0.4100

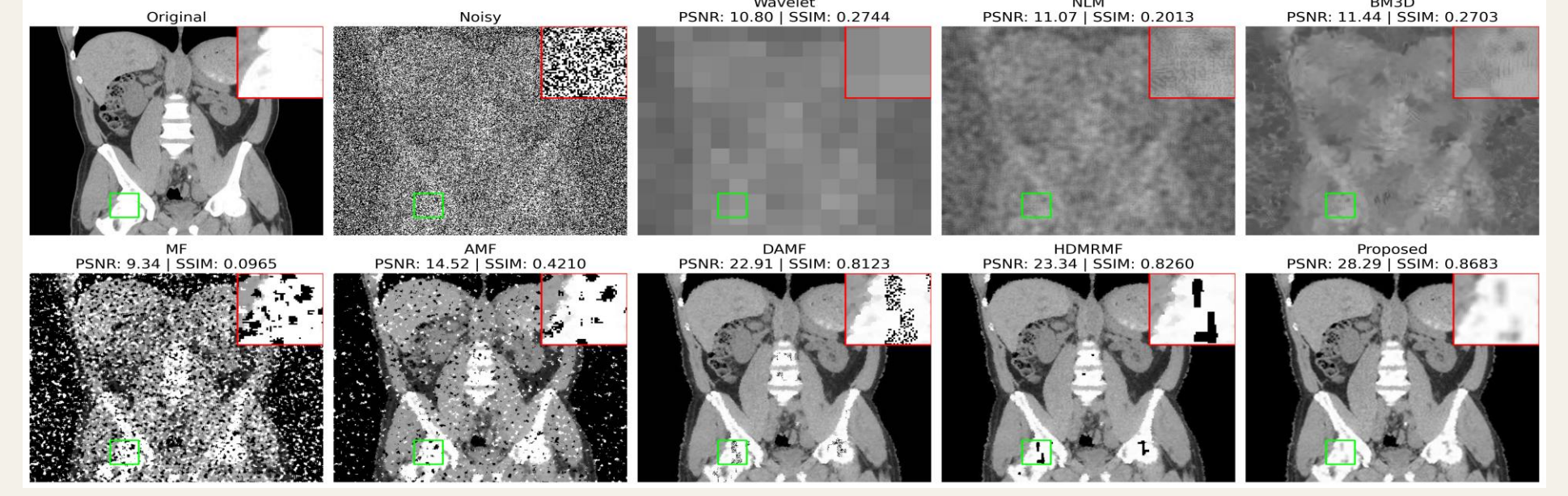


Fig. 6: Results for the Medical image under 0.8 noise
I-HDF: 28.29 dB | 0.8683
HDMRMF: 23.34 dB | 0.8260

- **Validated on Real Clinical Data:** Tested on real scans from the MedPix database, consistently outperforming competing baseline filters across all noise densities.
- **Preserves Anatomical Structures:** Reliably recovers soft contours, tissue gradients, and low-contrast regions without introducing false boundaries or distortion artifacts.
- **Practical Clinical Reliability:** Effectively eliminates heavy salt-and-pepper noise while maintaining vital structural contrast required for accurate medical imaging workflows.

Computational Time

Table 3: Comparison of Computational Time (Seconds)

Method	12STI	BSD68	Medical	Avg.
MF	0,002	0,001	0,002	0,002
AMF	3,901	2,305	3,625	3,277
DAMF	2,404	1,440	2,211	2,018
HDMRMF	8,168	4,947	7,747	6,954
I-HDF	2,058	1,289	1,958	1,768
BM3D	2,497	1,606	2,330	2,144
NLM	0,164	0,096	0,151	0,137
Wavelet	0,011	0,005	0,010	0,009

- 2nd fastest among median-based filters
- 4x faster than HDMRMF
- Outperforms DAMF and BM3D
- Lightweight CPU execution

Conclusion

- ✓ **Robust Hybrid Design:** Combines a modified adaptive median filter with approximate differential operators to effectively eliminate salt-and-pepper noise without sacrificing important image features.
- ✓ **Iterative LS Optimization:** Solves local overdetermined linear systems inside an iterative least-squares (LS) optimization framework to accurately recover fine details.
- ✓ **Effective Under Severe Noise:** Successfully restores sharp edges and fine textures even at 0.8 noise corruption avoiding the heavy blurring and patchy artifacts seen in standard filters.
- ✓ **Validated on Benchmark and Real Data:** Consistently outperforms standard baseline and advanced median-based filters in PSNR and SSIM across benchmark datasets (12STI, BSD68) and real clinical images from the MedPix database.
- ✓ **Fast and Practical Deployment:** Maintains a lightweight computational profile averaging 1.768 seconds on a standard CPU.
- ✓ **Future Work:** Extending the proposed framework to color (RGB) images and mixed noise models such as Gaussian and Rician noise.

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- [2] W. Wang, W. Du, Y. Li, and Z. Jia, "Advanced adaptive median filter for reducing salt-and-pepper noise in gpr data," IEEE Geoscience and Remote Sensing Letters, vol. 22, pp. 1–5, 2025.